Module 5: Membranes - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Describe how phospholipid structure creates a bilayer.
2. Predict which molecules cross membranes easily.
3. Compare passive and active transport.
4. Use concentration gradients to explain movement across membranes.
5. Predict osmosis outcomes in hypotonic, isotonic, and hypertonic conditions.

Topics

- Phospholipid bilayer structure
- Selective permeability
- Passive transport
- Active transport
- Osmosis and cell environments

Key Terms to Know

- **Phospholipid** - Molecule with a hydrophilic head and hydrophobic tails.
- **Bilayer** - Double layer of phospholipids forming a membrane.
- **Selective permeability** - Property of allowing some substances through more easily than others.
- **Diffusion** - Net movement from high to low concentration.
- **Osmosis** - Diffusion of water across a selectively permeable membrane.
- **Facilitated diffusion** - Passive transport through membrane proteins.
- **Active transport** - Energy-requiring movement against a gradient.
- **Tonicity** - Relative solute condition that predicts water movement.

Core Contents

1. The phospholipid bilayer forms a flexible boundary with hydrophilic and hydrophobic regions.
2. Selective permeability controls which materials cross easily and which require help.
3. Diffusion, osmosis, and facilitated diffusion move substances down gradients.
4. Active transport uses energy to move substances against gradients.
5. Tonicity predicts water movement and cell volume changes.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-05_viewing-life.md`

Generated Visuals

- Module 05: Membranes Evidence Map
- Module 05: Membranes Reasoning Sequence
- Module 05: Membranes Retrieval and Lab Check